

Home assignment 6**26.11.2024**

Problem 1. Consider two random variables X, Y with the joint distribution represented by the following table:

	$Y = -2$	$Y = 0$	$Y = 2$
$X = 1$	0.4	0.2	0.2
$X = 2$	0.1	0.1	0

- (a) Find their marginal distributions, $E(X)$, $E(Y)$, $\text{Var}(X)$, $\text{Var}(Y)$.
- (b) Are X and Y independent?
- (c) Find $\text{Cov}(X, Y)$ and $\rho(X, Y)$.
- (d) Find $\text{Cov}(2X - 5, 0.5Y - X + 3)$, $E(X \cdot (2Y^2 + 3))$.

Problem 2. Let X, Y, Z be random variables with expectations $\mu_x = 3$, $\mu_y = 1$, $\mu_z = 0$, and standard deviations $\sigma_x = 1$, $\sigma_y = \sigma_z = 2$. Suppose $\rho(X, Y) = 1/4$, and $X \perp\!\!\!\perp Z$, $Y \perp\!\!\!\perp Z$. Find $\text{Cov}(Y + 2Z + 1, 2X + Y - Z)$.

Problem 3. Suppose X takes on the values $-2, -1, 0, 1, 2$ with probability $1/5$ each, and let $Y = X^2$.

- (a) Find $\text{Cov}(X, Y)$.
- (b) Are X and Y independent?

Problem 4. Twenty coins are flipped. Let X be the number of heads on coins from 1 to 15, and Y be the number of heads on coins from 6 to 20. Find the correlation coefficient of X and Y .

Problem 5. Two teams are going to play two games. In each game one team wins and the other team loses, no draws are possible. The probability that one of the teams will win the first game is $1/2$. However, the team who lost the first game will be upset and will lose the second game with probability $3/4$.

Let X be the number of games (0, 1, or 2) won by the first team, and Y be the number of games won by the second team.

- (a) Find the distribution of X .
- (b) Find the joint distribution of X and Y .
- (c) Is the correlation coefficient $\rho(X, Y)$ positive, negative, or zero? Find it.